

Qingyu LIU

qliu91@jh.edu, <https://qingyuliu0521.github.io/>

EDUCATION

Johns Hopkins University | Baltimore, MD
M.S.E. in Electrical and Computer Engineering
GPA: 3.88/4.00

Aug. 2025 – Present

Shanghai Normal University | Shanghai, CHN
B.S. in Electronic Information Engineering (Outstanding Graduate)
GPA: 3.95/4.00 (Rank: 2/523, Top 0.5%)
TOEFL iBT: 104

Oct. 2020 – Jun. 2024

RESEARCH EXPERIENCE

X-Voice, Shanghai Jiao Tong University X-LANCE Lab
Co-First Author | Supervisor: Prof. Xie Chen

Oct. 2025 – May 2026

- Proposed a 0.4B-parameter multilingual flow-matching-based speech synthesis model trained on 420K hours of multilingual speech data, enabling zero-shot voice cloning across 30 languages.
- Designed a two-level language injection mechanism using FiLM modulation at the text level and LID embedding concatenation at the time level, reducing cross-lingual accent leakage.
- Adopted the supervised fine-tuning strategy from Cross-Lingual F5-TTS 2 with synthetic audio prompts, enabling zero-shot cross-lingual speech synthesis in 30 languages without requiring audio prompt transcripts.
- Led the development of the X-Voice repository: <https://github.com/sunnyxrxx/X-Voice>

Cross-Lingual F5-TTS 2, Shanghai Jiao Tong University X-LANCE Lab
Co-First Author | Supervisor: Prof. Xie Chen

Oct. 2025 – May 2026

- Proposed an SFT framework based on synthetic audio prompts, enabling zero-shot voice cloning without reference transcripts or forced alignment.
- Designed a text-conditioning strategy with learnable placeholders and EOS tokens to reduce the mismatch between pretraining and SFT-stage text conditions.
- Improved the speaking rate predictor from Cross-Lingual F5-TTS by introducing random silence injection, increasing robustness to leading and trailing silence in audio prompts.

Cross-Lingual F5-TTS, Shanghai Jiao Tong University X-LANCE Lab
First Author | Supervisor: Prof. Xie Chen | Accepted by ICASSP 2026

Mar. 2025 – Sep. 2025

- Developed a framework based on F5-TTS and MMS forced alignment, enabling cross-lingual zero-shot speech synthesis without requiring audio prompt transcripts.
- Designed and trained a speaking rate predictor to estimate synthesized speech duration from the speaking pace of the audio prompt.
- Deployed an interactive online demo on Hugging Face Space for public evaluation.

Infant Cry and Snoring Detection, Shanghai Normal University

Nov. 2023 – Jul. 2024

First Author | Supervisor: Prof. Yanhua Long | Accepted by Journal of the Audio Engineering Society

- Built and open-sourced the Infant Cry and Snoring Detection (ICSD) dataset for public research use.
- Proposed the Competitive CRNN+BEATs method, significantly improving detection performance on the ICSD dataset.
- Released the ICSD dataset and baseline code on GitHub.

INTERNSHIP EXPERIENCE

Johns Hopkins University, Center for Speech and Language Processing
Research Assistant | Supervisor: Prof. Berrak Sisman

Dec. 2025 – Present

Shanghai Jiao Tong University, X-LANCE Lab
Remote Research Assistant | Supervisor: Prof. Xie Chen

Aug. 2024 – Present

PUBLICATIONS

Note: * represents co-first authors; † represents corresponding authors.

- Qingyu Liu***, Rixi Xu*, Yushen Chen, Zhikang Niu, Haitao Li, Pengcheng Zhu, Bowen Zhang, Jian Zhao, Yunting Yang, Berrak Sisman, Kai Yu, Xie Chen†. "Cross-Lingual F5-TTS 2: Language-Agnostic Voice Cloning via Synthetic Prompts," 2026.
- Rixi Xu*, **Qingyu Liu***, Haitao Li, Yushen Chen, Zhikang Niu, Yunting Yang, Jian Zhao, Ke Li, Berrak Sisman, Qinyuan Cheng, Xipeng Qiu, Kai Yu, Xie Chen†. "X-Voice: Enabling Everyone to Speak 30 Languages via Zero-Shot Cross-Lingual Voice Cloning," *arXiv preprint arXiv:2605.05611*, 2026.

- **Qingyu Liu***, Yushen Chen, Zhikang Niu, Chunhui Wang, Yunting Yang, Bowen Zhang, Jian Zhao, Pengcheng Zhu, Kai Yu, Xie Chen[†]. "Cross-Lingual F5-TTS: Towards Language-Agnostic Voice Cloning and Speech Synthesis," in *ICASSP*, 2026.
- **Qingyu Liu***, Longfei Song, Dongxing Xu, Yanhua Long[†]. "ICSD: An Open-source Dataset for Infant Cry and Snoring Detection," *Journal of the Audio Engineering Society*.
- **Qingyu Liu***, Li Li, Yifan Zhou, Dongxing Xu, Yanhua Long[†]. "A Comparative Study of Deep Learning Methods for Infant Cry Detection," in *NCMMSC*, 2023.

AWARDS

Silver Award , 10th China International College Students "Internet+" Competition	Jun. 2024
Silver Award , 14th "Challenge Cup" College Students' Entrepreneurship Plan Competition	Jul. 2024
First Prize , 14th China National Undergraduate "Innovation, Creativity, and Entrepreneurship" Challenge	May 2024
Excellent Project Completion , National College Student Innovation Training Program	Nov. 2024